Yingrui Hou

Data Science & Scientific Data Engineering

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https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience in Python/C++ data analysis, TensorFlow/GPU numerical modelling and ML validation using real data. Combines statistical inference, reusable workflows and deep-learning model calibration. Adapts methods to new problems, coordinates analysis tasks and presents results in international teams. Targeting data science and scientific data engineering.

Core skills

- Programming & computing: Python, NumPy, pandas, scikit-learn, statsmodels; C/C++, ROOT/RooFit; TensorFlow and GPU numerical evaluation.
- Modelling: likelihood and mixture models, statistical inference, uncertainty analysis, CatBoost/gradient boosting, feature selection and distribution reweighting.
- Data workflows: reusable processing, Shell/YAML/CMake, Linux, shared-code review and reproducible analysis configurations.
- Model assessment: deep-learning calibration, image-recognition model validation, real-data performance checks, train/test diagnostics and bias correction.

Research experience

Postdoctoral Researcher | 2021–2026

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Developed reusable Python/C++/ROOT workflows linking data preparation, weak-signal extraction, likelihood fitting and validation when event-level ground truth was unavailable.
- Built CatBoost selection workflows with feature checks, train/test comparisons and threshold scans; saved models and configurations for reproducible application.
- Aligned simulated, control and target datasets through gradient-boosting reweighting and calibration; propagated residual modelling differences into uncertainty assessments.
- Used real data to evaluate and optimise the ECAL clustering model; investigated reconstruction discrepancies to guide calibration and software follow-up.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

TensorFlow/GPU scientific modelling

Constructed and evaluated complex-amplitude probability density models for a joint gamma measurement using TensorFlow and GPU acceleration. Applied numerical computing to a complex statistical model.

Deep-learning model calibration

Validated and calibrated a flavour-tagging model using real data, connecting learned-model outputs to subsequent physics analysis.

Image-recognition model performance

Used real data to check and optimise the model used for LHCb Run 3 ECAL clustering. Investigated reconstruction discrepancies and documented findings for calibration and software follow-up.

Machine learning under dataset shift

Built CatBoost classifiers with train/test checks, threshold studies and sample reweighting to align control and target data for subsequent inference.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

Research training applied to engineering

- Adaptability and learning: applied likelihood inference, CatBoost, TensorFlow/GPU and Geant4 to different research problems through literature, implementation and data checks.
- Research coordination: organised connected measurement, analysis and validation tasks with reusable components and documented assumptions.
- Presentation and teamwork: presented at GDR-InF 2022 and FCC/DRD France 2025, including on behalf of GRAiNITA; contributed to shared code and technical reviews.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- TensorFlow/GPU scientific modelling (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)
- Deep-learning model calibration (https://yingrui-hou.github.io/PersonalPage/index.html#work-9)
- Image-recognition model performance (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- Machine learning under dataset shift (https://yingrui-hou.github.io/PersonalPage/index.html#work-3)